

Analysis of 16S rRNA Microbial Community Sequencing Data

Fateema Bazzi

Introduction

Diet plays a crucial role in gut microbiome composition. In the paper by Filippo et. al. [1], researchers investigate how the fecal microbiota from Europe (EU) and a rural African village (BF) differ. Samples were retrieved and analyzed from 14 children belonging to the BF group and 15 children belonging to the EU group. Their findings showed significant differences in the gut microbiota between the two groups, suggesting the influence of diet on microbial diversity.

In this project I processed and analyzed the 16s ribosomal RNA (rRNA) sequencing data from the 29 samples using a standard microbiome data analysis workflow. The aim of this project was to highlight my technical competencies and biological interpretation skills.

Methods

I downloaded FASTQ files for each of the 29 samples. The data is publicly available on the European Nucleotide Archive [2]. I was provided the metadata for this project without any column names. Due to the ambiguity of many of the columns, I decided to only include the location grouping for each sample, EU or BF.

I assessed the initial quality of all 29 FASTQ files using FASTQC and MULTIQC to aggregate the results [3,4]. These quality control methods assess multiple aspects of the sequencing data. For 16s rRNA sequencing data, in my evaluation I prioritized per base sequence quality, per sequence quality scores, adapter content, sequence length distribution, and N content. Quality warnings in other aspects were to be expected due to the nature of the sequencing data and the fact that FASTQC does not distinguish between sequencing data types.

Next I looked at quality profiles for the samples, which illustrate the quality of the sequences along the bases, similar to the per base sequence quality in MULTIQC [5]. Looking at these I could make a decision as to where to trim the sequences based on where quality seemed to consistently drop across samples. I conducted filtration and trimming on the FASTQ files based on user specified criteria and returned the FASTQ files containing the trimmed reads that passed the filtering. In my use of the function, I adjusted two parameters from their defaults, truncLen and maxEE. The truncLen parameter specifies the truncation length, which I set to 250, and maxEE limits the amount of expected errors a read can have, and I set this to two for my workflow.

Once the data was filtered and trimmed, I learned the error rates of the data. I used these error rates, along with the reads, to infer the true sequence variants of the samples from the original unique reads. After sample inference, I was then able to construct the amplicon sequence variant (ASV) table. An ASV table “records the number of times each exact amplicon sequence variant was observed in each sample” [6]. The ASV table contains a row for every sample and a column for each unique sequence across all samples. The values within the table denote the number of occurrences of a unique sequence for a sample. From here, I removed chimeras from the ASV table. Chimeras are defined as sequences that “can be exactly reconstructed by combining a left-segment and a right-segment from two more abundant ‘parent’ sequences.”[6]

With the ASV table free of chimeras, I assigned taxonomy to the sequence variants. For my training set of reference sequences to make these classifications, I utilised the RDP trainset 19. RDP trainset 19 comes from the Ribosomal Database Project (RDP), which “provides the research community with aligned and annotated rRNA gene sequence data, along with tools to allow researchers to analyze their own rRNA gene sequences in the RDP framework” [7]. The authors of the paper where this data originated from also used the RDP reference training set [1], which motivated my decision to use it over others, although they used trainset 18, and I used the newer trainset 19. Taxonomy assignment implements a naive Bayesian classifier algorithm to return assigned taxonomies for each sequence.

To start multiple sequence alignment, I first built a DNASTringSet object using the sequences extracted from the ASV table, then I was able to perform multiple sequence alignment [8]. I set the anchor parameter to NA rather than its default 0.7 to prevent any bias using an anchor might cause. The small size of this dataset allowed for this without any detrimental slowdown. My next step was to build the phylogenetic tree [9]. I first computed pairwise distances using DNA sequences, then I was able to perform neighbor-joining tree estimation. I computed the likelihood of the tree given the sequence alignment and the tree. With the likelihood obtained I then optimized

the model parameters. I set the model to “GTR”, which is widely used for 16s rRNA sequencing data. I also set optInv and optGamma to TRUE in order to optimize the invariant proportion and gamma rate. Lastly, I set rearrangement to “stochastic” which performs a broad, slower rearrangement, but the small data size means the slow down is not too impactful and worth the improved results. Once the tree had been built and the parameters had been optimized, I could root it through midpoint rooting. With all the data I had collected at this point, I converted the ASV table into an operational taxonomic unit (OTU) table and took that along with the taxonomic data, tree, and metadata and combined them into a phyloseq object for downstream analyses [10].

Before doing any statistical analyses, I visualized sequencing depth, ASV richness, relative ASV abundance at the phylum and genus level, and ASV prevalence. I also visualized these features grouped by EU or BF samples for richness and abundance. Finally, I visualized the relationship between prevalence and abundance. Initially, I filtered the phyloseq object to remove any rare taxa present in only one sample. My intention was to filter out any mitochondrial or chloroplast sequences, but none were identified based on taxonomic assignment.

I assessed alpha diversity, which looks at the diversity of each sample. Rather than looking at just one measure, I looked at Observed, Shannon, Chao1, and Simpson. Due to the lack of singletons in the data the Chao1 index is invalid, so the Shannon index is the metric I focus on going forward. I used the Wilcoxon rank-sum test to assess whether the difference in alpha diversity was statistically significant. I then assessed beta diversity, which looks at the differences in group composition. I tested beta diversity using two different distance metrics, Bray-Curtis and weighted UniFrac [11,12]. With each distance metric, I performed ordination with the method of principle coordinate analysis and tested the statistical significance using PERMANOVA [13]. I also assessed the statistical significance of the dispersions using ANOVA.

To assess differential abundance, I used Analysis of Compositions of Microbiomes with Bias Correction (ANCOM-BC) [14]. ANCOM-BC determines which taxa abundances are significantly different between groups. Multiple testing correction was necessary to prevent an excess of false positives, and the method I used was Benjamini-Hochberg. I deemed taxa as significantly enriched if the absolute value of their log fold change was greater than one and if their q-value, which is the p-value after multiple testing correction, was less than 0.05. A positive log fold change indicated enrichment for the BF group, while a negative log fold change indicated enrichment for the EU group.

Results

The report returned from MULTIQC gave quality results within normal range for microbiome data. The per base sequence quality, along with the quality profiles, indicated that the quality of the reads dropped at around 250 base pairs, so that is where I decided to trim the reads.

At the initial construction of the ASV table, there were 2,384 sequences, all 250 base pairs long. After the identification and removal of 211 chimeras, there were 2,173 sequences. Approximately 75% of the original reads remained. Taxonomy assignment was able to assign most ASVs, and approximately 68% of the ASVs were assigned a genus.

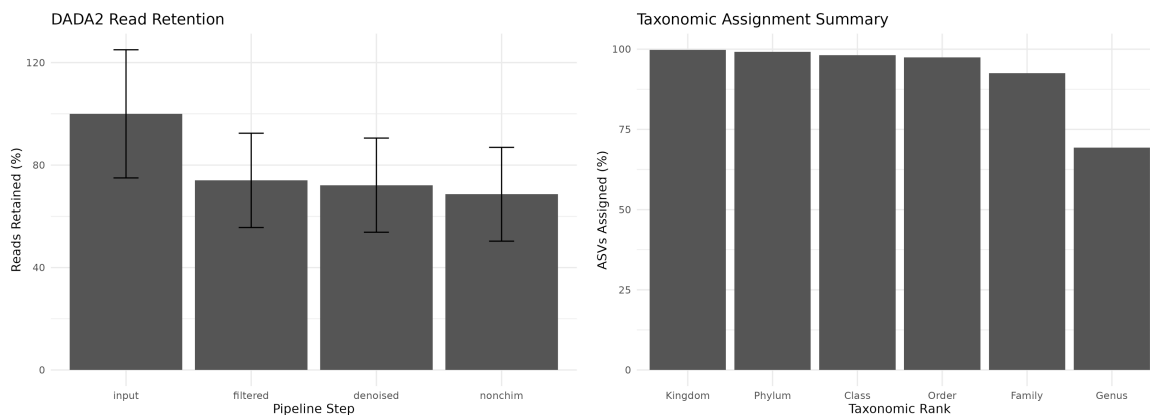


Figure 1: Bar plots representing the percent of reads retained after different steps in the workflow (left) and the percent of ASVs given a taxonomic assignment at different ranks (right).

Sequencing depth was visualized per sample and colored by the group the sample belonged to. None of the read counts were low enough to be considered an extreme outlier, with both groups represented at different depths. In terms of ASV richness, the EU group visually showed increased richness, but no statistical testing was done here, so it cannot be said to be significant.

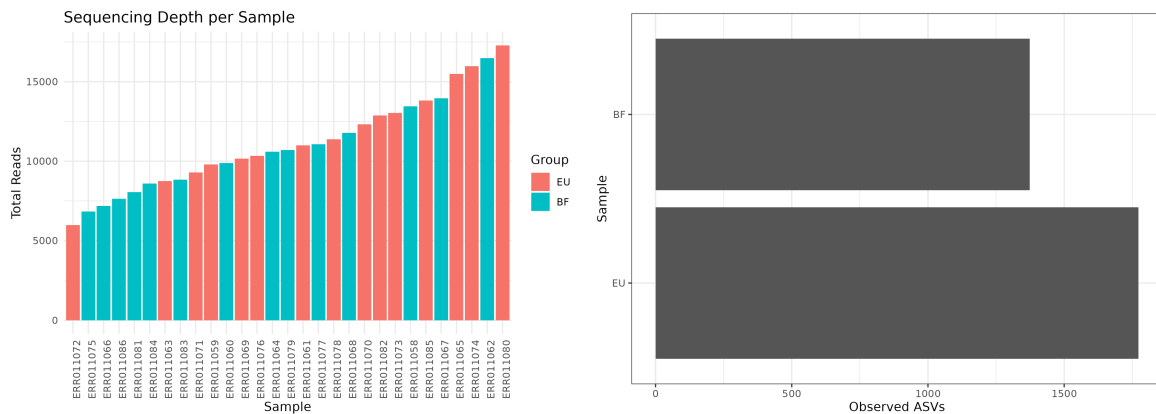


Figure 2: bar plots looking at the sequencing depth per sample, colored by group (left) and the ASV richness for each group (right).

I looked at relative abundance at the phylum and genus levels to see if the samples are dominated by a few ASVs. The samples as a whole were dominated by the phyla Bacillota and Bacteroidota. When grouped, Bacillota seemed to be more abundant in the EU group and Bacteroidota seemed to be more abundant in the BF group. At the genus level, Segatella seems to be more abundant in the BF group and Faecalibacterium seems to be more abundant in the EU group.

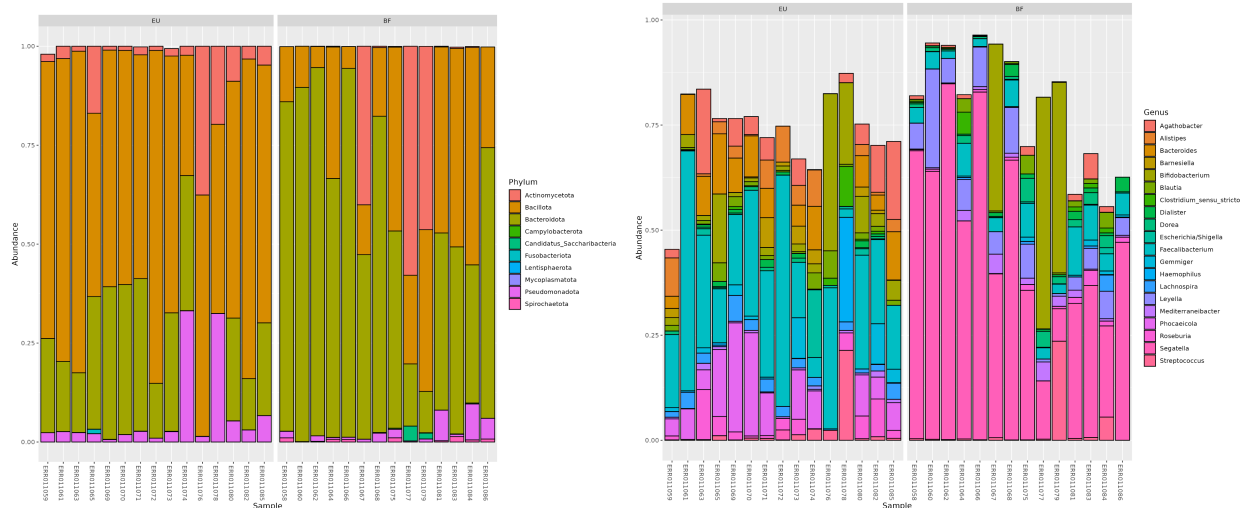


Figure 3: Relative abundance at the phylum (left) and genus (right) levels, separated by sample group.

Prevalence proves to be quite low, indicating that most ASVs were not detected in many samples and the dataset may contain many rare ASVs. Initially, I filtered out rare taxa, which significantly reduced the data from 2173 taxa to 522. In the end, I made the decision to continue working with the data without any rare taxa filtering.

When I assessed the Shannon index values between groups using the Wilcoxon rank-sum test, the results indicated a significantly higher diversity in the EU group compared to the BF group. After I calculated the beta diversity using both distance metrics, statistical testing indicated that with both options the two groups are significantly different from each other in their taxonomic composition.

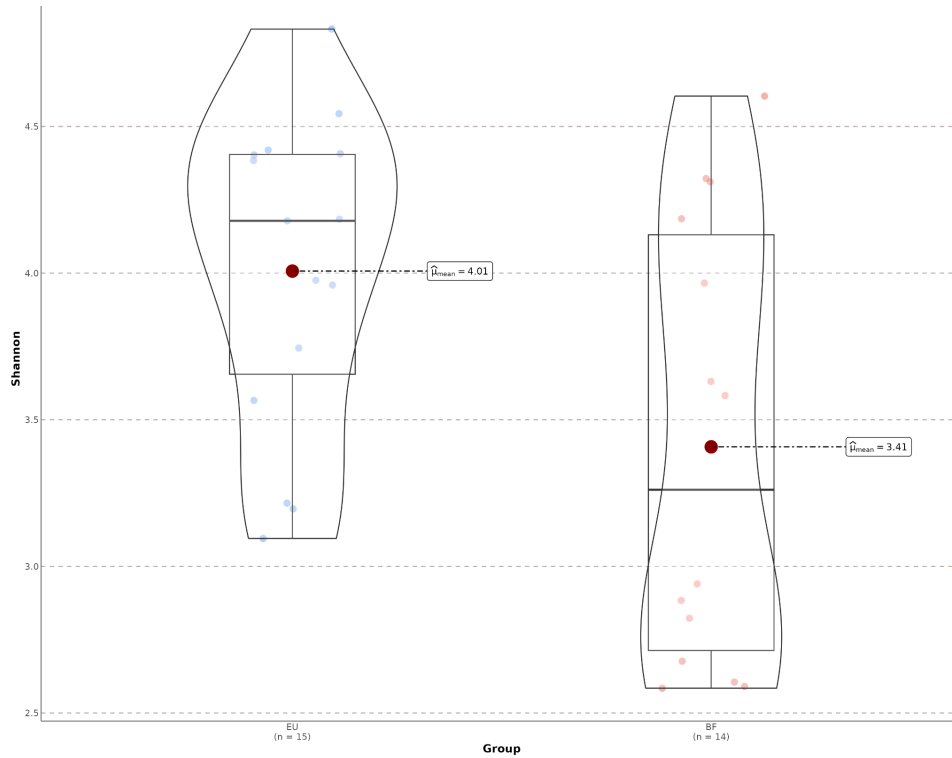


Figure 4: Violin plot comparing Shannon index between the sample groups.

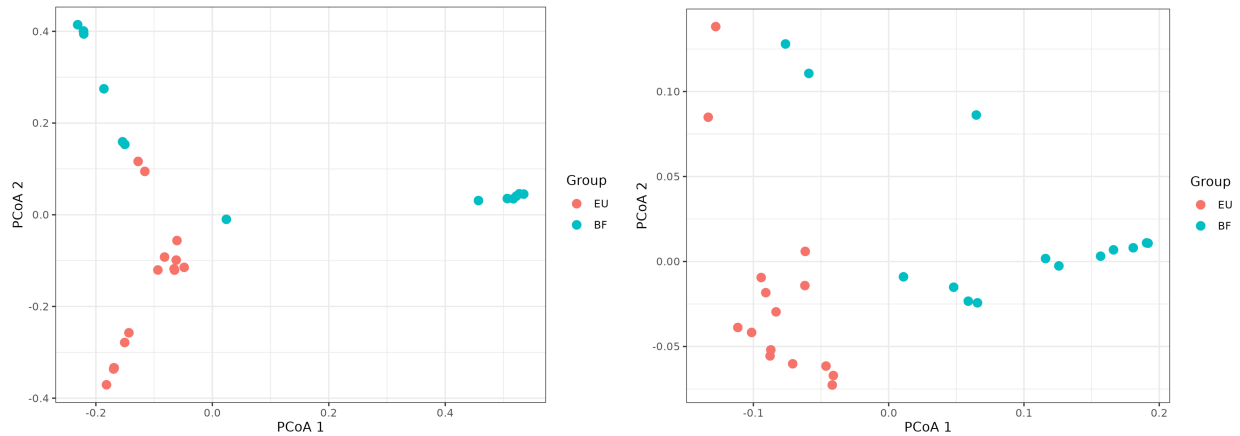


Figure 5: Ordination of the samples by principle coordinate analysis, Bray-Curtis (left), weighted UniFrac (right).

Differential abundance calculations and visualizations highlighted several taxa differentially enriched in one group or another. *Segatella*, *Leyella*, and *Prevotellamassilia* were most noticeably significantly differentially abundant in the BF group, while *Bacteroides* and *Phocaeicola* showed significance in the EU group. The heatmap clustered samples according to taxon abundance, with the exception of two BF samples, clearly separating the two groups.

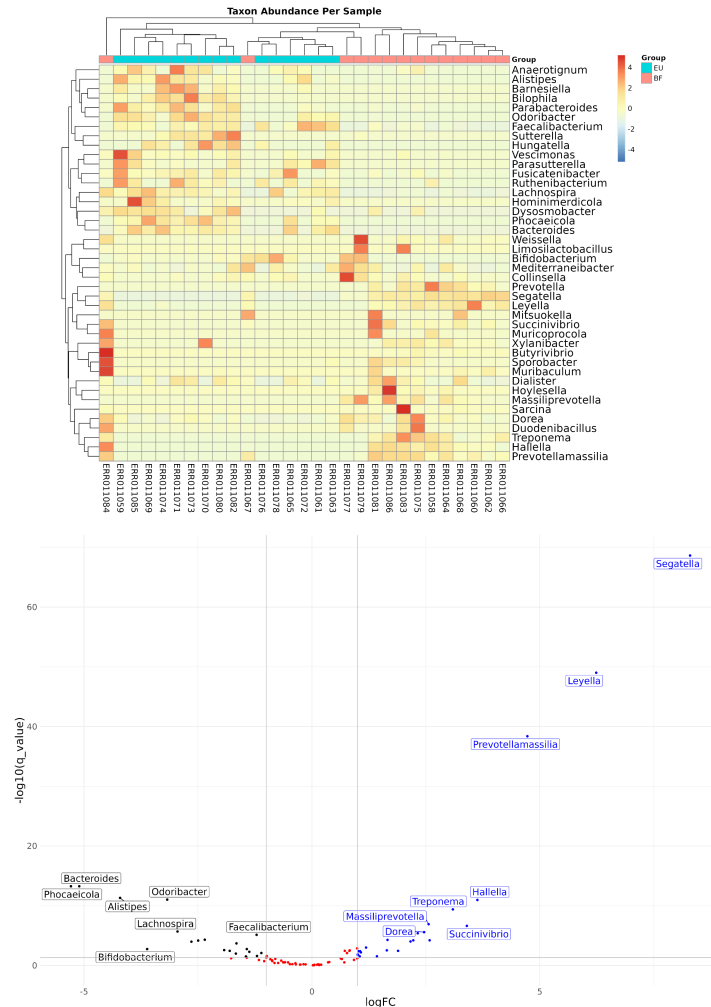


Figure 6: Heatmap (top) and volcano plot (bottom) illustrating differential abundance.

Discussion

The results of my analyses here indicate differences in the microbiota between children from Europe and children from the rural African village. It can be hypothesized that these differences are a result of the drastic differences in diet. The *Segatella* abundance in the BF group aligns with previous research indicating the prevalence of the genus in non-industrialized countries [15]. *Phocaeicola* increases in abundance after the introduction of solid foods, so I hypothesize that the abundance of this genus in the EU group is due to the increased duration of breast feeding within the BF group [16].

The original authors of this data, similar to my analysis, found that *Treponema* and *Prevotella* were differentially abundant in the BF group. However, the original paper stated that the BF samples displayed a higher microbial richness and biodiversity, whereas my results indicated the opposite. This contradiction could be the result of multiple factors, such as the difference in reference database used to assign taxonomy, or any different parameter choices along the analysis workflow.

While my workflow provides a thorough investigation of these microbial communities, several improvements could be made. The current denoising approach I employed used independent sample inference. Enabling pooled inference could increase sensitivity for detecting low-frequency sequence variants shared across multiple samples. Using the data filtered based on prevalence when calculating differential abundance testing could reduce the influence of extremely rare taxa and decrease the multiple testing burden. Finally, incorporating additional sample metadata could enable more comprehensive analyses and provide greater biological context for interpreting patterns.

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